Module 2 — Class 5 Assignment: Matplotlib Basics

Type-it-yourself exercise

UN AI/ML Mentorship

Module 2 — Class 5 Assignment: Matplotlib Basics

Topic: Matplotlib — bar, hist, scatter, box, subplots

Goal: Open a **fresh Google Colab notebook** and type the code yourself by following the steps below. Do NOT copy-paste from the working notebook. Typing it yourself is how you learn the syntax.

How to submit

1. Open Google Colab → File → New Notebook.
2. Type the code from each step below. Add a short `#` comment on EVERY cell so we know you understand it.
3. Save the notebook as `Module<X>_Class<Y>_<YourName>_Assignment.ipynb`.
4. Push it to your team repo at `module-<X>/class_<Y>/submissions/<YourName>/`.

Step 0 — Load the data

```
import pandas as pd
import matplotlib.pyplot as plt
url = 'https://raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d/master/data/Telco-Cu
df = pd.read_csv(url)
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce').fillna(0)
```

Step 1 — Bar chart — Contract types

What to do: Count how many customers have each Contract type and make a bar chart.

Type this in a new cell:

```
counts = df['Contract'].value_counts()
plt.figure(figsize=(7, 4))
plt.bar(counts.index, counts.values, color='teal')
plt.title('Customers per Contract type')
plt.ylabel('Number of customers')
plt.show()
```

Expected output:

```
A 3-bar chart: Month-to-month tallest, then Two year, then One year.
```

Step 2 — Histogram of tenure

What to do: Use plt.hist with 30 bins on the tenure column.

Type this in a new cell:

```
plt.figure(figsize=(8, 4))
plt.hist(df['tenure'], bins=30, color='coral', edgecolor='black')
plt.title('Distribution of customer tenure')
plt.xlabel('Tenure (months)')
plt.ylabel('Number of customers')
plt.show()
```

Expected output:

```
A histogram with peaks at the very low end (new customers) and very high end (long-time custome
```

Step 3 — Scatter — MonthlyCharges vs TotalCharges

What to do: Make a scatter plot of MonthlyCharges (x) vs TotalCharges (y).

Type this in a new cell:

```
plt.figure(figsize=(8, 5))
plt.scatter(df['MonthlyCharges'], df['TotalCharges'], alpha=0.3, s=10, color='navy')
plt.title('Monthly vs Total Charges')
plt.xlabel('Monthly Charges')
plt.ylabel('Total Charges')
plt.show()
```

Expected output:

```
A scatter showing positive correlation - higher monthly charges = higher total charges.
```

Step 4 — Box plot — MonthlyCharges by Contract

What to do: Compare MonthlyCharges across the 3 Contract types using a box plot.

Type this in a new cell:

```
groups = [df[df['Contract'] == c]['MonthlyCharges'] for c in df['Contract'].unique()]
labels = df['Contract'].unique()
plt.figure(figsize=(7, 5))
plt.boxplot(groups, labels=labels)
plt.title('Monthly Charges by Contract')
plt.ylabel('Monthly Charges')
plt.show()
```

Expected output:

```
3 box plots side by side, showing the spread of monthly charges per contract type.
```

Step 5 — Subplots — 1 row, 3 charts side by side

What to do: Create a 1x3 figure with bar, hist, and scatter together.

Type this in a new cell:

```
fig, ax = plt.subplots(1, 3, figsize=(15, 4))
ax[0].bar(df['Churn'].value_counts().index, df['Churn'].value_counts().values, color='gray')
ax[0].set_title('Churn count')
ax[1].hist(df['MonthlyCharges'], bins=30, color='orange')
ax[1].set_title('Monthly Charges')
ax[2].scatter(df['tenure'], df['TotalCharges'], alpha=0.2, s=5)
ax[2].set_title('Tenure vs Total Charges')
plt.tight_layout()
plt.show()
```

Expected output:

```
A wide figure with 3 charts side by side.
```

Checklist before you submit

- ☐ All cells run without error
- ☐ Every code cell has a `#` comment in your own words
- ☐ Outputs are visible (you saved AFTER running)
- ☐ File name is correct: `Module<X>_Class<Y>_<YourName>_Assignment.ipynb`

Deadline: before next class.